

neering theory exists to answer these questions. Packet sizes are standardised for different data networks by different professional bodies. Table I has details on the same.

Packet switching is used in all networks, as shown in Table I. It is used to transport conventional data. Over time, there emerges a logical need to have a switching technique for all services, voice, video and data as multimedia services. In packet switching, packets are variable in sizes and relatively long. Variable packet sizes require higher processing time at transmitter (for making of frames with formation of overheads) as well as receiver (for removing overheads, rearranging packets in order, as packets may go out of order for transport over different available parallel paths, error checking, etc) sides.

Voice and video are time-sensitive services. Accordingly, packet switching was modified as asynchronous transfer mode (ATM) cell switching to make it applicable to voice and video, besides data. In ATM cell switching, packet size is fixed at 53 bytes with 48 bytes of data and five bytes of overhead. Due to a fixed packet size, node processing delay is minimised.

Further, the short size of a cell in ATM gives the benefit of availing more parallel paths and faster transport. This makes ATM (often known as very fast packet switching) an appropriate switching technique for multimedia services.

Packet switching is implemented in two modes: virtual circuit services (VCS) for huge data and datagram service (DS) for low data.

In VCS, logical virtual connection is pre-established for data transport between source and destination. All data of the session between source and destination follow the virtual link. But unlike telephony, the pre-established link is sharable.

In DS, the packet moves independently over available links between source and destination.

The motivation behind data networking was to connect geographically-distributed computing resources (standalone computers, terminals, printers, scanners, etc) through communication links (that may be wires,

cables, optical-fibre lines, wireless links, etc) with a view to meaningful exchange of information. There are two primary goals: resource sharing and load sharing, so that costly printers, software and computing devices through networking can be shared. This results in cost-sharing among multiple users connected in the network, thereby making networking a cost-effective solution.

In case of huge load, standalone computers, when connected in a network, may use other computers for sharing the load. Thus, reliability and flexibility of implementing computing enhances through networking.

Data networks are classified in various ways. One of the ways is based on the routing mechanism of packets. In shared media networks (basically local area networks, or LANs), routing is made by media access control (MAC). In switched networks (metropolitan area networks, or MANs, and wide area networks, or WANs), packets are routed through intermediate nodes, often known as switching nodes.

Another means of classification of networks is based on geographical coverage areas of networks. Accordingly, networks are classified as LAN, MAN and WAN. See Table II for details.

The US military began the research and development of data networks in 1950s. But in 1969, researchers from California University, Los Angeles and Santa Barbara, Stanford Research Institute and University of Utah came up with a successful data network having a 50kbps connection link of four nodes in the four university locations. The network was implemented under Advanced Research Project Network (ARPANET), USA.

In 1973, Bob Metcalfe launched the Ethernet—IEEE 802.3-based LAN—as a discovery. This was a small step in data networking, for today's big leap in the Internet. As a natural process of technological development, thereafter, MAN and WAN gave way to X.25 technology. **EFY**

This article is dedicated to Leonard Kleinrock, who conceived the great idea of the Internet in his paper titled "Information flow in large communication nets" in the year 1961.



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Plot No. EL-1, MIDC Electronic Zone,
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Tel : 0091-22-2767 3300 (Board), 2767 3311-16 (Sales)
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